

Structural Descriptors for Metal–Organic Frameworks: Multifractal Analysis of Framework Connectivity for Simulation-Free Screening

23CSE498 – Project Phase 2 (Panel Review 1) | Team Number: [XX]

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Guide: [Guide Name], Designation, Department of CSE

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0. Problem Context — Why This Project Exists

The gap between the number of candidate materials and our ability to evaluate them

The materials-science problem

- **Metal–organic frameworks (MOFs)** are crystalline solids built by joining metal clusters (*nodes*) to organic molecules (*linkers*).
- They are extremely porous — internal surface areas above 3,000 m²/g — making them candidates for gas storage, carbon capture and separation.
- **Hundreds of thousands** of MOFs are known or hypothesised.

The computational bottleneck

- Evaluating one MOF by molecular simulation (GCMC) takes **hours**.
- Exhaustive screening of the design space is **not feasible**.

Research Question

Can a **purely structural, simulation-free descriptor** — computed from a framework's *connectivity* in seconds — group MOFs into meaningful families, so a new candidate can be triaged **without** expensive calculation?

Value Proposition

Not a *replacement* for simulation — a **filter** that decides **what to simulate**. Reducing 10⁵ candidates to 10² is the contribution.

1. Methodology & Solution Design

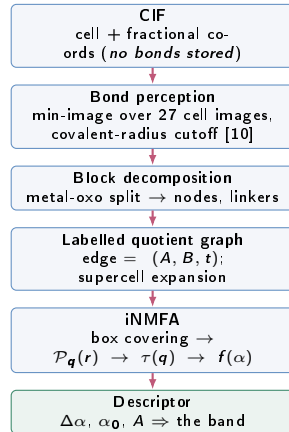
Rubric: Workflow, Architecture, Algorithms/Mathematical Model & Evaluation Metrics (5 Marks)

Theoretical foundation. A MOF is a **periodic graph**. Collapse each building block to a vertex and each inter-block bond to an edge; the chemistry reduces to pure connectivity, which the properties follow.

Core algorithms

- **Decomposition:** MOFid metal-oxo algorithm [1] — PBC-aware bond perception, connected-component block extraction.
- **Graph construction:** *labelled quotient graph* — edges carry a lattice translation t , so the infinite crystal is represented exactly.
- **Descriptor:** influential-node multifractal analysis (iNMFA) [2].

Dataset. 77 frameworks queried live from MOFX-DB [7], with published PLD/LCD pore diameters. Supercells of 3,400–6,100 atoms.



Validation metric: published coordination numbers — an independent ground truth the graph must reproduce.

1b. Mathematical Model

The descriptor, defined

Step 1 — the measure. For each influential node i (top 10% by degree) and box radius r , let $M_i(r)$ be the number of vertices within r steps:

$$p_i(r) = M_i(r)/N$$

Step 2 — the partition function, reweighted by a distortion factor q :

$$\mathcal{P}_q(r) = \sum_{i \in I} p_i(r)^q$$

Step 3 — the mass exponent.

$$\tau(q) = \frac{\ln \mathcal{P}_q(r)}{\ln(r/r_N)}$$

Step 4 — Legendre transform.

$$\alpha(q) = \frac{d\tau}{dq}, \quad f(\alpha) = q\alpha(q) - \tau(q)$$

Interpreting the spectrum

Quantity	Physical meaning
α	local growth rate around one block
$f(\alpha)$	how much of the framework shares it
α_0	the <i>typical</i> dimension (apex)
$\Delta\alpha$	heterogeneity of the pore network
A	asymmetry — dense-vs-open dominance

Structural constraint used as a correctness test

$f(\alpha)$ must be an inverted parabola peaking at $q = 0$, where $f(\alpha_0) = D_0$. Any implementation not producing this shape is wrong — we used it as a test, and it caught a real defect (Slide 3).

Evaluation metrics: coordination-number agreement with published crystallography; spectrum shape (% peaking at $q = 0$); seed reproducibility; partial correlation against pore diameter.

2. Implementation Progress ($\approx 60\%$) & Module Functionality

Rubric: Core Modules Implemented, Functional & Integrated (8 Marks)

Module / Component	Current Implementation State	Status
Data ingestion (code_01-02)	CIF parsing, symmetry expansion, PBC bond perception over 27 cell images	Functional
Decomposition (code_03-04)	<code>is_metal()</code> classification; metal-oxo, single-node, all-node splits	Functional
Topology export (code_05-07)	Centroid simplification, Systre .cgd writer, MOFid/MOFkey assembly	Functional
Network analysis (code_09)	Labelled quotient graph, degree/eigenvector/ betweenness centrality, Laplacian	Functional
Spectrum (code_10-11)	iNMFA (box-covering) and NMFA (box-growing); τ , α , $f(\alpha)$, $\Delta\alpha$, A	Functional
77-framework study	Live MOFX-DB query, spectra for all 77, band + confound analysis	Functional
Interactive web platform	18-section site; full pipeline runs live in-browser on any uploaded CIF	Integrated
Size-confound correction	Fit-window normalisation / size-matched comparison	In Progress
Experimental-set validation	Re-query CoRE MOF [9] for genuine metal diversity	Planned

Verification. Two independent implementations — Python (reference) and JavaScript (browser) — agree on every reported value ($\tau(0) = -2.6551$

on HKUST-1 in both). Automated test suites cover the decomposition, the network layer, the band analysis and the site structure. Code:

github.com/Radha-Krishna-19/fyp

2b. Results Obtained So Far

Every number computed from the live pipeline — nothing transcribed

Validation — the graph is correct

	Node c.n.	Published	Net
HKUST-1	4	(4, 3)	tbo
MOF-5	6	(6, 2)	p cu
ZIF-8	4	(4, 2)	sod
UiO-66	12	(12, 2)	f cu

All four match. A naive construction gives UiO-66 = 6.

Defects make influence measurable

In a perfect crystal every block is symmetry-equivalent $\Rightarrow$ only **2** distinct centrality values $\Rightarrow$ “top- k ” is a **tie**. Remove one linker: $2 \rightarrow 4$ distinct values, and $\lambda_2 : 1.44 \rightarrow 1.19$ — the weakening, quantified.

The band — 77 frameworks

Spectra peaking at $q = 0$	77 / 77
$\Delta\alpha$ range	1.016 – 1.410
Interquartile band	1.112 – 1.250
Asymmetry A	–1.51 to –1.04 (all negative)

A tight band with a consistent asymmetry sign.

What the band does *not* yet mean

$\Delta\alpha$ vs pore diameter, raw	–0.497
controlling for graph size	+0.101
$\Delta\alpha$ vs graph size, ctrl. pore	+0.621

R^2 : size alone / +pore 0.533 / 0.538

Pore size adds **+0.005**. $\Delta\alpha$ tracks **supercell size** — a computational parameter, not chemistry.

3. Technical Knowledge & Design Decisions

Rubric: Understanding of Methodology, Implementation & Design Choices (5 Marks)

Why a graph representation? MOF properties follow *connectivity*: gas moves through the pore network, and the pore network is the void left by the wiring. An **unweighted** graph deliberately discards element identity — what survives is **pure architecture**, which transfers across chemistries.

Key design decisions

- **Labelled quotient graph over a naive graph.** Without translation labels, UiO-66's 12-connected cluster reports as 6 — or degenerates into a star.
- **Supercell expansion.** A primitive cell has diameter ~ 2 , too small for a power-law fit. Expansion is *exact* because edges carry translations.
- **Box-growing (deterministic) validated against box-covering (stochastic).**
- **Defect simulation** to break symmetry — physically justified by [15].

Defect found by our own shape test

The spectra were not parabolic. Cause: $\tau(q)$ was extracted with a **free-intercept** least-squares slope, but [2] defines it as a **ratio** $\ln \mathcal{P}_q / \ln(r/r_N)$ — a fit **through the origin**. At $q = 0$: every term is $p_i^0 = 1$, so $\mathcal{P}_0 = |I|$ is **constant in r** . A flat line has slope $0 \Rightarrow \tau(0) = 0 \Rightarrow f(\alpha_0) = 0$ — **the peak is deleted**.

τ definition	$\tau(0)$	$f(\alpha_0)$	parabola
free intercept	0.0000	0.0000	no
paper (ratio)	−2.5669	+2.5669	yes

Trade-offs. MIN_CELL_LENGTH/MAX_ATOMS trade scaling range against $O(N^2)$ memory — and are the source of the size confound we identified.

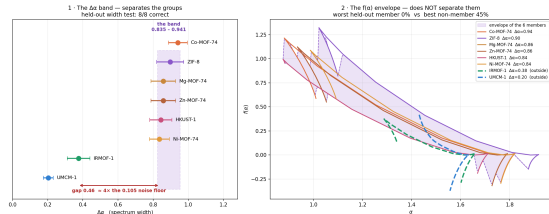
Influential-node selection is ambiguous in a perfect crystal (all degrees tie); we report the ambiguity rather than hide it.

3b. Deliverable — Interactive Analysis Platform

An 18-section technical site; the full pipeline executes in-browser

What it does

- Runs the **complete decomposition live** on any uploaded .cif — no server, no installation.
- Steps through all nine pipeline modules with the **real source code beside a live 3D model** of the structure at that stage.
- Computes the network analysis and the multifractal spectrum **in the browser**, cross-validated against the Python reference.
- A **graph-theory primer** (Section 10) defines every term — vertex, degree, quotient graph, Laplacian, λ_2 , defect — before it is used.
- Every source file is readable by clicking its name anywhere on the page.



The $\Delta\alpha$ band and the $f(\alpha)$ spectra, generated by the analysis notebook.

4. Technical Enrichment Activity Choices

Professional Online Certification Details (Individual Mandatory: 10–20 Hours)

S.No	Student Name	Chosen MOOC / Course Title	Platform	Hours
1	P. M. Radha Krishna	Social and Economic Networks: Models and Analysis	Coursera (Stanford)	20 Hrs
2	P. Kshitij Varma	Introduction to Complex Network Analysis	NPTEL	20 Hrs
3	G. Rohith Abhinav	Chemoinformatics / Computational Materials Science	NPTEL	16 Hrs
4	J. Keerthi Sree	Data Visualization with Python	Coursera (IBM)	15 Hrs
5	Isha	Machine Learning for Materials Informatics	edX / Coursera	15 Hrs

Rationale for these selections. Each maps to a component this project actually uses: **network analysis** (the graph representation and centrality measures), **complex networks** (fractal and multifractal characterisation), **chemoinformatics** (CIF handling, bond perception, MOF structure), **data visualisation** (the interactive platform), and **materials informatics** (descriptor-based screening — the project's stated goal).

Requirement: each student completes one approved 10–20 hour certification from a recognised platform relevant to the project.

5. Remaining Work & Timeline

Concrete engineering tasks, not open research questions

Priority 1 — Remove the size confound

- Normalise the fit window per structure so $\Delta\alpha$ stops growing with graph size; **or** compare only frameworks at matched graph size.
- Re-test the pore-diameter correlation afterwards. *This decides whether the band carries chemical meaning.*

Priority 2 — A genuinely diverse dataset

- Re-query CoRE MOF 2019 [9] with per-record database verification.
- Current set is 77 hMOF entries [8], **all Zn_4O** — hypothetical and single-metal, so no cross-metal claim is possible.

Priority 3 — Validate against measured properties

- Correlate the descriptor with published adsorption data to test whether it predicts anything, rather than merely describing structure.

Position of the work

The **machinery is complete and verified**: the decomposition, the periodic graph, the centrality analysis and the spectrum all work, are cross-validated across two implementations, and reproduce published crystallography.

What remains is establishing **whether the descriptor means anything physically** — and we have identified precisely what blocks that and how to remove it.

On reporting negative results

We report the size confound rather than the raw -0.497 correlation. A band built on an uncontrolled size effect would not survive scrutiny; identifying and quantifying it is the more defensible scientific position.

References I



References II



Thank You

Interactive platform, source code and full documentation:
github.com/Radha-Krishna-19/fyp